

SEC P vs NP log 2026-04-27

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-27 03:18 UTC

SEC P vs NP — daily log 2026-04-27

8 cycles recorded

Tropical Affine Rigidity: Vanishing Discrepancy iff Single-Atom Fourier Concentration

- **Verdict:** INCONCLUSIVE
- **Bridge:** Tropical Fourier Analysis (Tropical Geometry / Idempotent (Maslov) Harmonic Analysis) $\times$ Affine-Function Recognition Decision Problem (membership in the AC^0 -decidable class of degree-1 tropical polynomials, equivalently the zero-discrepancy promise problem)
- **Recorded:** 2026-04-27 00:22 UTC
- **Entry ID:** 81928f9c440e

Statement

Let f be a TropicalPolynomial over a finite domain $D = \{-N, \dots, N\} \subset \mathbb{Z}$ (min-plus semiring). Let $F = \text{FourierTransform}(f)$ denote its TropicalFourierTransform (Legendre-Fenchel transform: $F(y) = \min_{\{x \in D\}} \{f(x) - xy\}$ over a finite slope grid Y), and let $m(f) = \text{MinimalFourierCoefficient}(f) = \min_{\{y \in Y\}} F(y)$. Then $\text{DiscrepancyCalculation}(f) = 0$ if and only if f is tropically affine, i.e., there exist a in Y and b in R with $f(x) = ax + b$ for all x in D ; moreover, in this affine case the support $\{y : F(y) = m(f)\}$ is the singleton $\{a\}$ and $m(f) = -b$. Equivalently, $\text{DiscrepancyMeasure}(f) > 0$ iff the level set $\text{argmin}_y F(y)$ has cardinality unequal to 1 in the regular position OR F attains its minimum $-b$ at exactly the unique slope a only when f is affine — formally: $\text{Discrepancy}(f) = 0 \Leftrightarrow |\{y \in Y : F(y) = m(f)\}| = 1 \wedge f(x) \equiv (\text{argmin}_y F(y)) * x - m(f)$.

Rationale

This sub-conjecture sharpens axiom A3 (DiscrepancyMeasure bounded by $\max |\text{Fourier coefficient}|$) at its extremal saturation point. A3 says discrepancy is controlled from above by Fourier magnitudes; the natural converse extremal case is when discrepancy collapses to zero: this should force the Legendre-Fenchel transform to be a singular Dirac-like object on the slope grid, since affine functions are the unique ‘tropically uniform’ (curvature-free) elements. The conjecture additionally invokes A2 (invertibility of TropicalFourierTransform on a privileged subclass): affine polynomials are precisely the fixed atoms of the involution from the existing Legendre-Fenchel sub-conjecture, so the singleton-support claim gives a constructive certificate that those polynomials sit at the boundary of the invertibility class. If true, this yields a linear-time complexity-theoretic decision procedure for affineness via discrepancy evaluation.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1207.2443v2] Tropical Teichmüller and Siegel spaces - [1204.4578v1] Complexity of tropical and min-plus linear prevarieties - [2006.04823v3] Quantum Legendre-Fenchel Transform - [0710.0377v1] Idempotent and tropical mathematics and problems of mathematical physics (Volume I) - [0709.4119v1] Idempotent and tropical mathematics and problems of mathematical physics (Volume II)

Empirical Test

- exit code: 0, elapsed: 5.79s

```
"instances_tested": 600, "conjecture_holds": true, "counterexample": ""}}
{"seed": 23, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 37, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 53, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 71, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
RESULT: INCONCLUSIVE not enough seeds supported
```

Judge reasoning

The test's own RESULT line explicitly states 'INCONCLUSIVE not enough seeds supported', and the suspiciously constant 0.6667 metric across all seeds suggests a tautological aggregate check rather than genuine verification of the biconditional's adversarial failure modes. | next: Rewrite the test to explicitly construct piecewise-linear f with multiple slopes tying for $\arg\min F(y)$, verify singleton-support on affine instances, and report per-seed Pearson correlation on perturbed-affine stress tes

Continued Fraction Depth of Spectral Ratio Bounds Discrepancy

- **Verdict:** INCONCLUSIVE
- **Bridge:** Diophantine approximation (continued-fraction expansions and partial-quotient growth, in the spirit of Khinchin and Gauss-Kuzmin metric theory) $\times$ Discrepancy-based randomized two-party communication complexity lower bounds for ± 1 sign matrices
- **Recorded:** 2026-04-27 00:31 UTC
- **Entry ID:** 71ce01b16c1e

Statement

Let $M \in \{-1, +1\}^{N \times N}$ with $N=2^n$ have singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_N$, and let $\rho(M)=\sigma_2/\sigma_1 \in [0,1]$. Let $(a_1, a_2, \dots, a_K)$ be the simple continued-fraction expansion of $\rho(M)$ truncated at $K=\lceil \log_2 N \rceil$ partial quotients (computed via the Euclidean algorithm on a $4N$ -digit decimal truncation of ρ), and define $S(M)=\sum_{i=1}^K \log(1+a_i)$. Then for every $n \geq 3$ and every ± 1 sign matrix M of size $2^{n \times 2}n$, $\text{disc}(M) \cdot N \geq c \cdot \sigma_1(M) / (1 + S(M))$ for an absolute constant $c=0.05$, where $\text{disc}(M)$ is the combinatorial rectangle discrepancy of M .

Rationale

Highly structured matrices (Hadamard, Sylvester, AND/XOR-lifted indicators) have spectra that are integer or low-height rational, forcing σ_2/σ_1 to be exactly 0, 1, or a fraction with tiny partial quotients — yielding small $S(M)$ and hence the well-known large $\text{disc} \cdot N \approx \sigma_1$. Generic matrices satisfy Gauss-Kuzmin statistics with partial quotients distributed like $1/(k(k+1))$ and $S(M)=\Theta(\log N)$, exactly compensating their $\sigma_1=\Theta(\sqrt{N})$ into $\text{disc} \cdot N=\Theta(\sqrt{N})$. The Diophantine-approximation lens has, to my knowledge, never been applied to the spectral side of the discrepancy method, even though Khinchin-style metric theory is the natural quantitative refinement of 'how rational is the spectral ratio'.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1210.1461v1] A Scalable CUR Matrix Decomposition Algorithm: Lower Time Complexity and Tighter Bound - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [2209.08318v3] Hausdorff dimension of sets with restricted, slowly growing partial quotients in semi-regular continued fractions - [1510.00068v2] On the complete solution of the general quintic using Rogers-Ramanujan continued fraction

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5248d06.py", line 137, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5248d06.py", line 113, in run_trial
    M = M_type()
        ^^^^^^^
```

TypeError: run_trial.<locals>.<lambda>() missing 1 required positional argument: '_'

Judge reasoning

The test crashed with a `TypeError` (`lambda` missing positional argument `'_'`) before any trials produced data, so neither the support nor falsification criteria can be evaluated. | next: Fix the `M_type` `lambda` signature in `run_trial` so all four ensembles can be instantiated uniformly, then rerun the 140-triple sweep.

Young-Flattening Rank Gap Lower-Bounds Padded-Permanent Border Rank

- **Verdict:** INCONCLUSIVE
- **Bridge:** Geometric Complexity Theory via Young flattenings (Landsberg-Ottaviani equivariant rank obstructions) $\times$ Border rank of the padded $m \times m$ permanent tensor Per_m viewed as a slice of det_n
- **Recorded:** 2026-04-27 01:05 UTC
- **Entry ID:** 0e80f2d2c50a

Statement

Let Per_m be the $m \times m$ permanent and det_n its $n \times n$ padded determinant embedding ($n \geq m^2$). Let $\text{YF}_k(T)$ denote the rank of the Young flattening of tensor T indexed by the partition $(k, 1^{\{m-k\}})$ acting on the symmetric m -th power of $C^{\{m,2\}}$. Conjecture: for all $2 \leq m \leq 6$ and all $1 \leq k \leq m-1$, the equivariant rank gap $\delta(m,k) := \text{YF}_k(\text{det}_n \text{ viewed as } \text{Per}_m\text{-padding}) - \text{YF}_k(\text{Per}_m)$ satisfies $\delta(m,k) \geq c \cdot \binom{m}{k} \cdot (n - m^2 + 1)$ with absolute constant $c = 1/2$, and consequently the border rank lower bound $\text{brk}(\text{Per}_m) \geq \max_k \text{YF}_k(\text{Per}_m) / \binom{m-1}{k-1}$ is tight to within a factor of m .

Rationale

Young flattenings are the only currently known equivariant obstructions that have produced super-additive border-rank lower bounds for permanents (Landsberg-Michalek), yet no quantitative law

connects their rank gaps to the padding parameter n ; pinning down $\delta(m,k)$ would convert orbit-closure containment questions into a finite linear-algebraic check. Because Young flattenings are computed by listing semistandard Young tableaux and writing an explicit polynomial-sized matrix of monomial coefficients, the obstruction is constructive and respects the GL_n^2 -symmetry that defeats natural-proofs and algebrization barriers. A clean quantitative law here would directly sharpen Mulmuley-Sohoni's program for separating VP from VNP via Per vs Det.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [1993636.1993704] Geometric complexity theory and tensor rank - [s2:1011.1350] Geometric Complexity Theory and Tensor Rank

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 69, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 50, in run_trial
    YF_k_det_n = young_flattening_rank(det_n, m, 1)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 39, in young_flatten
    return len(gaussian_elimination(M))
               ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 24, in gaussian_elim
    factor = -A[j][i] / A[i][i]
              ~^^^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an `IndexError` in `gaussian_elimination` before producing any $\delta(m,k,n)$ data, so the pre-registered support condition could not be evaluated. Per Rule 2, a crash yields INCONCLUSIVE. | next: Fix the `gaussian_elimination` routine (handle zero-pivot/short-row cases, e.g., add partial pivoting and bounds checks) and rerun the deterministic sweep over $m \in \{2,3,4\}$, $1 \leq k \leq m-1$, $n \in \{m^2, \dots, m^2+4\}$.

Pebble-Cost Gadget Lifts Decision Tree Depth to KW-Game Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Black-white pebbling games on rooted DAGs (Cook-Sethi pebbling cost) $\times$ Karchmer-Wigderson communication complexity on lifted query problems
- **Recorded:** 2026-04-27 01:38 UTC
- **Entry ID:** 52cebd2ffe62

Statement

Let f be a partial Boolean function on m bits and let G_f be the canonical AND/OR DAG of its minimum-depth decision tree T_f , with black-white pebbling cost $p(G_f)$. Define the lifted relation $f \circ g$ where g is the 2-bit indexing gadget `IND_2` on $b=2$ address bits, and let $KW(f \circ g)$ denote the deterministic KW-game length on $f \circ g$. Then for every f with $|T_f| \leq 2^m$ and $m \leq 12$, $KW(f \circ g) =$

$\text{depth}(T_f) + p(G_f) \pm 1$, i.e. the lifted KW length equals the tree depth plus the pebbling cost of its DAG up to additive 1.

Rationale

Raz-McKenzie style lifts add a multiplicative gadget overhead, but the *exact* additive correction has never been pinned to a classical DAG invariant; pebbling cost measures exactly the simultaneous-memory needed to simulate the tree as a circuit, which is precisely the resource the KW-prover must spend to track gadget answers. If correct, this gives a tight (not asymptotic) lifting identity computable on tiny instances, sharper than known $O(\text{depth} \cdot \log b)$ bounds. It also explains why IND_2 already suffices for many separations: the gadget cost is absorbed by pebbling rather than by an extra log factor.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_db2bed9c.py", line 116, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_db2bed9c.py", line 92, in run_trial
    m = random.randint(3, 5) if m == 6 else m
                                   ^
```

UnboundLocalError: cannot access local variable 'm' where it is not associated with a value

Judge reasoning

The test crashed with an UnboundLocalError before producing any data, so the pre-registered support condition (≥ 200 samples with $|k-d-p| \leq 1$ for 100%) was never evaluated. No counterexample was observed either, so neither support nor falsification is established. | next: Fix the variable-scoping bug in run_trial (initialize m before the conditional reassignment) and rerun the full sweep over $m \in \{3, \dots, 12\}$ to obtain the required ≥ 200 samples.

Mealy Automaticity Lower-Bounds Truth-Table Formula Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Finite-state transducer minimization / 2-automaticity of binary sequences (Allouche-Shallit automaticity over the alphabet $\{0,1\}$, distinct from any compression-based complexity) $\times$ Meta-complexity: minimum $\{\text{AND,OR,NOT}\}$ -formula leaf count $L(f)$ of a truth table, used as a tractable MCSP proxy
- **Recorded:** 2026-04-27 02:14 UTC
- **Entry ID:** 33ac9cdf1529

Statement

For an n -variable Boolean function f with truth table $T_f \in \{0,1\}^{2^n}$, let $A(f)$ be the number of states of the minimum complete DFA over alphabet $\{0,1\}$ that accepts $L_f = \{ \text{bin}_n(k) : T_f[k] = 1 \}$, where $\text{bin}_n(k)$ is the n -bit MSB-first binary expansion of k . Let $L(f)$ be the leaf-count of the smallest $\{\text{AND,OR,NOT}\}$ -formula computing f over the literals $x_1, \dots, x_n, \neg x_1, \dots, \neg x_n$. Conjecture:

for every $n \geq 2$ and every f , $A(f) \leq (n+2) \cdot L(f) + 2$; equivalently, $\log_2 A(f) \leq \log_2 L(f) + \log_2(n+2) + o(1)$. A single Boolean function violating this bound refutes the conjecture.

Rationale

Mealy/DFA state count is a transducer-theoretic measure that has been intensively studied for infinite automatic sequences but almost never tied directly to MCSP-style formula minimization on truth tables; small formulas should yield few distinguishable prefix-evaluation states, suggesting at most a linear-in- n blow-up from $L(f)$ to $A(f)$. This bridge converts an MCSP question into a poly-time-computable transducer invariant (Hopcroft minimization), giving a falsifiable surrogate without needing to solve MCSP itself. If the bound fails, the offending function exposes a ‘state-rich, formula-cheap’ object — a new natural property candidate that would not relativize, fitting the META_COMPLEXITY safe-direction profile.

Novelty

- Judge: NOVEL over 10 arXiv hits

Top hits consulted: - [2003.09703v1] Variance function of boolean additive convolution - [1112.2313v2] QBF-Based Boolean Function Bi-Decomposition - [2402.02935v4] Nuclear mass table in deformed relativistic Hartree-Bogoliubov theory in continuum, II: Even- Z nuclei - [2006.04708v2] On three conjectures of P. Barry - [1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 139, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 118, in run_trial
    A_f = hopcroft_minimization(set(range(2**n)), truth_table_to_dfa(truth_table, n)[1], set())[0]
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 95, in truth_table_to_dfa
    transitions[s]['0'] = get_next_state(s, '0')
                        ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 89, in get_next_state
    for i in range(len(state)):
                  ^^^^^^^^^
```

TypeError: object of type 'int' has no len()

Judge reasoning

The test crashed with a `TypeError` in `truth_table_to_dfa` before any function was evaluated, so no $A(f)$ or $L(f)$ data was produced and the pre-registered support condition cannot be checked. | next: Fix the DFA construction (state representation type bug in `get_next_state`) and rerun the enumeration/sampling protocol over $n \in \{2, \dots, 8\}$.

Tropical Positive-Scalar Homogeneity of MinimalFourierCoefficient

- **Verdict:** INCONCLUSIVE

- **Bridge:** TROPICAL_FOURIER_ANALYSIS (tropical geometry $\times$ Fourier analysis over the max-plus semiring) $\times$ Min-plus circuit complexity / tropical circuit lower bounds (a complexity-theoretic measure governing shortest-path and dynamic-programming circuits)
- **Recorded:** 2026-04-27 02:28 UTC
- **Entry ID:** 6be01f5e483c

Statement

Let f be a TropicalPolynomial on a finite grid $X \subset \mathbb{R}^d$, and let $\lambda > 0$. Define the tropically rescaled polynomial $g = \lambda \odot f$, i.e., the pointwise rescaling $g(x) = \lambda \cdot f(x)$ (tropical exponentiation by λ). Let $F = \text{TropicalFourierTransform}(f)$ and $G = \text{TropicalFourierTransform}(g)$ be their tropical Fourier coefficients (the Legendre–Fenchel-type transforms guaranteed invertible by axiom A2 on the relevant class). Then $\text{MinimalFourierCoefficient}(g) = \lambda \cdot \text{MinimalFourierCoefficient}(f)$, and $\text{DiscrepancyCalculation}(g) = \lambda \cdot \text{DiscrepancyCalculation}(f)$. In particular, the bound from axiom A3, $\text{DiscrepancyMeasure}(h) \leq \max_k |\hat{F}_h(k)|$, scales homogeneously of degree 1 under positive tropical rescaling, so the inequality is preserved exactly under $\lambda > 0$ (no slack opens or closes).

Rationale

This conjecture directly tests axiom A3 ($\text{DiscrepancyMeasure} \leq \max \text{Fourier coefficient}$) jointly with axiom A2 (invertibility of TropicalFourierTransform on the chosen class). Because tropical multiplication by a positive scalar corresponds to ordinary multiplication, and the Legendre–Fenchel transform satisfies $(\lambda \cdot f)(y) = \lambda \cdot f(y/\lambda)$, both extremes of f^* (its minimum and the discrepancy it controls) must scale by exactly λ . Verifying this 1-homogeneity is a clean sanity check that the framework’s notion of MinimalFourierCoefficient behaves as a genuine ‘norm-like’ invariant compatible with the tropical semiring’s scalar action, and that the A3 bound is not an artifact of a particular scale (relevant for transferring lower bounds in tropical/min-plus circuit complexity, where rescaling weights is a standard reduction).

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [1010.5964v1] Quadratic discrete Fourier transform and mutually unbiased bases - [1207.2443v2] Tropical Teichmüller and Siegel spaces - [1204.4578v1] Complexity of tropical and min-plus linear prevarieties - [1406.3065v2] Lower Bounds for Tropical Circuits and Dynamic Programs - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity?

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_21db269d.py", line 110, in <module>
    result = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_21db269d.py", line 71, in run_trial
    f = [max([random.randint(-10, 10) * x[i] + random.randint(-10, 10) for i in range(d)]) for _ in range(5000)]
            ^
```

NameError: name 'x' is not defined

Judge reasoning

The test crashed with a NameError (‘x’ is not defined’) before any trials executed, so no data was produced to evaluate the pre-registered support or falsification conditions. | next: Fix the test harness by properly defining the grid points x within the polynomial generation (e.g., evaluate the affine forms over each grid point), then rerun the 5000 trials.

Permanent-Variance of NW Design Matrix Predicts PRG Bias

- **Verdict:** INCONCLUSIVE
- **Bridge:** Combinatorics of 0/1 matrix permanents (Ryser-formula variance over random row-sign perturbations) $\times$ Nisan-Wigderson pseudorandom generator output bias against small DNF distinguishers
- **Recorded:** 2026-04-27 02:42 UTC
- **Entry ID:** 92f05cc72075

Statement

Let D be an (ℓ, a) -NW design on n variables encoded as an $m \times n$ 0/1 incidence matrix M (rows = subsets S_i , columns = universe). Define $V(M) := \log_2(1 + \text{Var}_{\{\text{eps} \in \{+/-1\}^n\}}[\text{perm}(M_{\text{eps}})]) / m$, where M_{eps} is M with column j scaled by eps_j and perm is the 0/1 permanent on the $(m \times m)$ submatrix induced by the m heaviest columns. We conjecture that for every hard truth table $f : \{0,1\}^\ell \rightarrow \{0,1\}$ of formula-leaf-count $L(f) \geq 2^{\ell/3}$, the maximum bias of $\text{NW}_{\{D,f\}}$ against width- w DNF distinguishers satisfies $\text{bias}(\text{NW}_{\{D,f\}}, w) \leq 2^{-c * V(M) * m / w}$ for an absolute constant $c > 0$, so $V(M)$ is a constructive lower bound certificate for PRG quality.

Rationale

Permanent variance over sign perturbations measures how ‘spread’ the design’s row-overlap pattern is, which is exactly the combinatorial quantity controlling pairwise-intersection cancellation in the NW hybrid argument; existing analyses use only $|S_i \cap S_j| \leq a$, but the variance aggregates higher-order interference. Linking permanent fluctuation (a 0/1 matrix invariant computable in $O(m * 2^m)$ by Ryser) to NW bias is, to our knowledge, unstudied and sidesteps natural-proofs because the invariant depends on the design only, not on f ’s truth table.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1311.6531v3] Brains and pseudorandom generators - [2412.05161v1] DNF: Unconditional 4D Generation with Dictionary-based Neural Fields - [2404.12647v1] Simple constructions of linear-depth t -designs and pseudorandom unitaries - [2003.09703v1] Variance function of boolean additive convolution - [2012.01667v2] Computing the matrix fractional power with the double exponential formula

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_08ccc640.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_08ccc640.py", line 66, in run_trial
    perm_var = permanent([[M_heaviest[i][j] * M_heaviest[j][i] for j in range(m)] for i in range(m)])
                      ~~~~~^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an `IndexError` before producing any data, so no support fraction or violation ratio could be computed against the pre-registered criterion. | next: Fix the indexing bug in the permanent computation (the `M_heaviest` submatrix access uses `M_heaviest[j][i]` where `j` ranges over `m` but rows are indexed differently) and rerun the trials.

Cycle-Space Cocycle Imbalance Lower-Bounds Resolution Width of Tseitin

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic graph theory: $\text{GF}(2)$ cycle space and edge-boundary cocycle imbalance of expander graphs $\times$ Resolution refutation width of Tseitin formulas on bounded-degree expanders
- **Recorded:** 2026-04-27 03:18 UTC
- **Entry ID:** 09c50eb1a365

Statement

Let $G=(V,E)$ be a connected 3-regular graph and let $T(G,c)$ be the Tseitin formula for an odd charge $c:V \rightarrow \{0,1\}$. Define the cocycle imbalance $I(G) = \min \text{ over nonempty } S \text{ subset } V \text{ of } |E(S, V \setminus S)| / \min(|S|, |V \setminus S|)$, and let $w(T(G,c))$ denote the minimum width of any resolution refutation of $T(G,c)$. The conjecture is the linear lower bound $w(T(G,c)) \geq \text{floor}(I(G) * \min(|S|, |V \setminus S|) / 2) + 1$, where S is the cocycle-imbalance minimizer; equivalently, resolution width is at least half the minimum odd-charge edge-boundary plus one.

Rationale

Ben-Sasson-Wigderson tie resolution width to expansion of the underlying constraint hypergraph, and Urquhart's classical Tseitin lower bounds are driven by edge-boundary expansion of small vertex sets. Reformulating expansion as a $\text{GF}(2)$ cocycle imbalance (the dual of cycle-space sparsity) gives a coordinate-free invariant that should sharpen the constant in width lower bounds, and falsification on a single 3-regular graph would refute a clean algebraic strengthening of the BSW bound.

Novelty

- Judge: NOVEL over 7 arXiv hits

Top hits consulted: - [1411.3958v3] Edge-state enhanced transport in a 2-dimensional quantum walk - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [0804.4433v1] SPACE: the SPectroscopic All-sky Cosmic Explorer - [2405.02292v1] ALOHA 2: An Enhanced Low-Cost Hardware for Bimanual Teleoperation

Empirical Test

- exit code: 124, elapsed: 180.08s

TIMEOUT after 180s

Judge reasoning

The test timed out after 180s (returncode=124) without producing any RESULT line, so the pre-registered support condition over ≥ 210 instances was never evaluated. | next: Re-run with a higher time budget and/or a faster width-bounded resolution solver (e.g., restrict to $n \leq 14$ first, parallelize per-instance, and cache $I(G)$ computations) to actually collect the 210 data points.